Fine-Tuning a 600M NLLB-200 Model for Low-Resource Spanish-Quechua Machine Translation

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***Abstract*—Quechua, a low-resource agglutinative language, is spoken by more than 3 million people in Peru. It still faces challenges with large-scale, open-domain, and diverse dataset for preservation and use in neural machine translation. In this work, we fine-tune the 600M distilled version of NLLB-200 on 102k Spanish-Quechua sentence pairs from the SomosNLP 2022 dataset. The model achieves a chrF++ score of 42.68 on the test set of this dataset, and 51.88 on the official AmericasNLP 2024 development set. However, the latter result is affected by domain overlap between the training and evaluation data, as both come from related sources. The findings suggest that a carefully fine-tuned 600M model can be competitive with larger 1.3B models, while also highlighting the need for more diverse dataset and morphology-aware tokenization for Quechua.**

***Index Terms*—Quechua, neural machine translation, low-resource language, NLLB-200, fine-tuning, multilingual language models**

I. Introduction

Quechua is the most widely spoken indigenous language family in the Americas, with the Peruvian variety alone accounting for more than three million speakers [1]. It is also one of the languages where the gap between sociolinguistic importance and digital representation is widest: despite the size of its speaker population, Quechua remains a low-resource language in almost every practical sense relevant to natural language processing, lacking large, clean, openly licensed parallel corpora comparable to those available for English, Spanish, or other major world languages.

Part of the difficulty is linguistic. Quechua is agglutinative and highly synthetic, meaning that grammatical information that would require several separate words in Spanish or English, such as tense, aspect, evidentiality, negation, person, and number, is instead encoded by chaining suffixes onto a single root. A single Quechua verb form can therefore correspond to an entire clause in Spanish, and the same root can take on dozens of surface forms depending on which suffixes are attached and in what order. This is compounded by dialectal variation: Quechua is not a single uniform language but a family of related varieties spoken across Peru, Bolivia, Ecuador, and parts of Colombia and Argentina, with differences in phonology, morphology, and orthography that are large enough to affect mutual intelligibility in some cases. Any machine translation system trained for "Quechua" is, in practice, trained for one variety, usually whichever one dominates the available data, which in our case is the Ayacucho-Chanka variety written in the standardized alphabet promoted by Peru's Ministry of Education.

The other part of the difficulty is the simple scarcity of data. Even the larger Quechua corpora that exist today are an order of magnitude or more smaller than what is considered a normal training set for a high-resource language pair, and much of what is available was assembled opportunistically, from religious texts, government bulletins, community-contributed translations, or hackathon projects, rather than curated specifically for machine translation. This matters because data quality issues, misalignments, inconsistent orthography, duplicate or near-duplicate sentences, untranslated segments, do not average out the way they might in a corpus of hundreds of millions of sentences; in a 100k-sentence corpus, every flawed pair is a measurable fraction of the training signal.

Multilingual translation models such as NLLB-200 [2] were designed precisely to address this kind of scarcity, by training a single massive model across more than 200 languages so that low-resource languages can benefit from shared representations learned from higher-resource ones. Quechua is included in NLLB-200's language inventory, but as several AmericasNLP shared task papers have shown, the base checkpoint performs quite poorly on Quechua out of the box, and some amount of task-specific fine-tuning is necessary before the model produces usable output [3]. What remains an open, practical question, especially for researchers and small teams without access to industrial compute budgets, is how much fine-tuning is actually needed, and at what model scale, to get a translation system that is good enough to be useful for the kinds of applications low-resource language communities actually care about, things like education tools, content access, and language documentation.

This paper investigates that question with a deliberately modest and reproducible setup. We take the 600M distilled variant of NLLB-200, roughly half the parameter count of the 1.3B model most commonly used in prior Quechua-focused

work, and fine-tune it end-to-end on the SomosNLP 2022 Spanish-Quechua dataset [4], a community-built corpus of about 102,000 sentence pairs. We deliberately use this dataset exactly as released, without additional cleaning, deduplication, filtering, or orthographic normalization, because one of the goals of this work is to measure what a straightforward, low-effort fine-tuning pipeline can achieve with data as it actually exists in the wild, rather than under idealized, curated conditions. We then evaluate the fine-tuned model on two different test sets: the held-out test split of the same SomosNLP dataset, and the official AmericasNLP 2024 development set [3], which was constructed independently by a different team for a shared task benchmark. Reporting both numbers side by side, rather than just the more flattering one, lets us examine not only how well the model fits the distribution it was trained on, but how its apparent performance shifts, for better or worse, when it is tested against data it has never seen, assembled for an entirely different purpose.

We want to be explicit from the outset about what this paper is, and is not, claiming. This is not an attempt to build a state-of-the-art, production-grade translation system, and the numbers reported here should not be read as directly comparable to commercial systems or to carefully curated academic benchmarks where data cleaning, filtering, and quality control were part of the pipeline. Because the training data was used in raw form, the model inevitably absorbs whatever noise, inconsistency, and misalignment already existed in the SomosNLP corpus, and this caps how far the results can be pushed regardless of model size or training budget. Our contribution is narrower and, we think, still useful: a transparent account of what a 600M model can achieve with realistic, unglamorous data and limited compute, and an honest discussion of where the resulting numbers should and should not be trusted.

Our contributions can be summarized as follows:

- We fine-tune NLLB-200-distilled-600M on the raw, unfiltered SomosNLP Spanish-Quechua corpus and report chrF++ and BLEU on its own test split, obtaining 42.68 chrF++ and 17.67 BLEU.
- We evaluate the same model, with no further adaptation, on the official AmericasNLP 2024 development set, reaching 51.88 chrF++, a number that on its face exceeds previously published results on that benchmark, including a 3.3B parameter model fine-tuned with LoRA adapters.
- We show through qualitative error analysis and a discussion of dataset provenance that this apparently strong result is substantially inflated by topical, stylistic, and possibly source-level overlap between SomosNLP and the AmericasNLP data, and we argue that it should not be read as evidence that our system is state of the art or industry-competitive.
- We identify subword tokenization, which is not adapted to Quechua's agglutinative morphology, and the lack of any data cleaning, as the two most likely bottlenecks limiting further improvement, and we lay out concrete next steps for addressing both.

The rest of the paper is organized as follows. Section II gives some linguistic background on Quechua relevant to the modeling choices discussed later. Section III reviews prior work on Spanish-Quechua and Indigenous-language NMT more broadly. Section IV describes the dataset, the fine-tuning setup, and our evaluation protocol. Section V presents the quantitative results on both test sets. Section VI discusses the qualitative behavior of the model and the limitations of comparing across datasets. Section VII lays out the limitations of this study explicitly. Section VIII concludes with directions for future work.

II. Background: Quechua and the Challenges of Agglutination

Before discussing the experimental setup, it is worth spending a short section on why Quechua's structure matters so directly for the modeling choices in this paper, rather than treating it as a generic "low-resource language" the way it is sometimes framed in the broader NLP literature.

Quechua verb morphology is built around a root followed by an ordered sequence of suffixes drawn from a fixed inventory, each contributing a specific piece of grammatical meaning. Object and subject person markers, tense, aspect, evidentiality (whether the speaker witnessed an event directly, inferred it, or was told about it), and a range of discourse-level particles can all stack onto a single verb. The order of these suffixes is largely fixed, but the number that can appear on any given word is not, which means the space of possible surface word forms is enormous compared to the number of underlying roots. This is the defining property of agglutinative languages, and it interacts badly with the kind of subword tokenization, byte-pair encoding or SentencePiece, that almost all modern NMT systems, including NLLB-200, are built on. These tokenizers learn their vocabulary from corpus statistics, which means common roots and common short suffixes may be well represented, but the long, low-frequency tail of complex inflected forms gets fragmented into pieces that do not correspond to actual morpheme boundaries. The model then has to learn, largely from scratch and from limited data, how these fragments recombine to produce valid Quechua words, a harder learning problem than the equivalent task for a more isolating or fusional language like Spanish.

A second relevant property is orthographic variability. Quechua did not have a single standardized writing system until relatively recently, and even now, different institutions, regions, and individual writers use somewhat different conventions, particularly around vowel inventories (three vowels in the standardized system versus five in some traditional or Spanish-influenced spellings) and the representation of certain consonant clusters. A dataset compiled from multiple sources, as SomosNLP's corpus is, will likely contain a mix of these conventions, which adds another layer of surface-level variation that the model has to absorb without any explicit signal that two differently spelled forms are the same underlying word.

Finally, Quechua is really a family of mutually related but not fully mutually intelligible varieties. Cusco-Collao Quechua, Ayacucho-Chanka Quechua, and Quechua varieties spoken in Ecuador or Bolivia differ enough in vocabulary, phonology, and some grammatical features that a model trained primarily on one variety, as ours is, should not be expected to generalize well to the others. NLLB-200's `quy_Latn` language code corresponds specifically to Ayacucho Quechua, and this is the variety that dominates both the SomosNLP corpus and, to our knowledge, most of the AmericasNLP Quechua data, which is itself a relevant point when interpreting why scores on the two datasets end up closer together than one might expect from two independently constructed corpora.

III. Related Work

A. Multilingual Models for Low-Resource Translation

NLLB-200 [2] represented a substantial step forward for low-resource machine translation by training a single model across more than 200 languages with shared parameters, allowing transfer from higher-resource languages to lower-resource ones in the same family or typological neighborhood. Quechua is included among the supported languages, but as multiple shared-task papers have documented, the off-the-shelf model performs quite poorly on it without further adaptation, which has made fine-tuning NLLB checkpoints on whatever Quechua parallel data is available a fairly standard recipe in recent work, including ours.

B. The AmericasNLP Shared Task Series

The AmericasNLP workshop has run an annual shared task on machine translation into Indigenous languages of the Americas since 2021, and it has become the closest thing this subfield has to a standard benchmark. The first edition introduced an open machine translation track covering ten languages and released a from-scratch baseline trained on the shared corpus; the REPU CS team's Spanish-Quechua submission [5] was among the earlier entries to explore targeted data augmentation for this specific pair. That same year, the Helsinki team [6] submitted a multilingual baseline system covering several language pairs simultaneously, and this system, along with a refined version submitted by the Sheffield team in 2023 [7], has since been adopted by the organizers as the standard reference baseline against which later submissions are compared.

The most directly relevant prior work for this paper is the BSC submission to the 2024 edition of the shared task [3], which is also the origin of the official development set we use for our second evaluation. The BSC team fine-tuned NLLB-200 at both 1.3B and 3.3B parameter scale, using LoRA adapters for the larger model specifically to keep the compute budget manageable, training only 14.2% of the total parameters while reportedly matching the performance of full fine-tuning. Their reported chrF++ scores on the Spanish-to-Quechua dev set, in the mid-thirties, were the highest in that edition of the competition and outperformed both the Helsinki and Sheffield baselines as well as a Google Translate comparison point. Our work takes a complementary angle: rather than scaling the model up and using parameter-efficient adapters to manage cost, we use a smaller, fully fine-tuned model on a larger, if less curated, training corpus, to see how far data volume alone can compensate for reduced model capacity.

C. Tokenization and Morphology in Indigenous-Language MT

A separate line of work has focused less on model scale and more on how text is segmented before it reaches the model. Ortega et al. [8] studied neural machine translation specifically for polysynthetic languages and argued, with Quechua and related languages as motivating examples, that generic subword tokenization schemes systematically mishandle morpheme boundaries in agglutinative and polysynthetic languages, and that this mismatch is a meaningful contributor to translation errors independent of dataset size or model architecture. This is a point we return to directly in Section VI, since several of the qualitative errors we observe in our own outputs are consistent with this diagnosis.

D. Other Spanish-Quechua Translation Efforts

Outside the shared task ecosystem, smaller-scale academic projects have also tackled this language pair. Ren [9] explored transfer learning approaches for Spanish-to-Quechua translation as part of a graduate course project, fine-tuning sequence-to-sequence models pretrained on higher-resource languages and demonstrating measurable improvements over training a comparable model from scratch, though working at a smaller scale than the pretrained multilingual models used in this paper. More recently, the UCSP team's submission to the 2025 edition of AmericasNLP [10] continued refining the recipe with updated data and baselines, suggesting that the field is still actively converging on best practices for this language pair rather than having settled on one. Our work is best read as one more data point in this ongoing process, specifically addressing the question of how much a smaller, less expensive model can achieve when given more, if rougher, training data.

IV. Dataset and Methodology

A. Dataset

We use the Spanish-Quechua dataset released by SomosNLP during their 2022 hackathon [4], distributed through the Hugging Face Hub. It contains approximately 102,000 aligned sentence pairs, split by the original authors into training, validation, and test partitions, with the test split alone containing 12,843 examples. The corpus was assembled from a mix of community-contributed translations and openly available bilingual text, predominantly in the Ayacucho-Chanka variety of Quechua written in standardized orthography, though as discussed in Section II, some internal inconsistency in spelling conventions should be expected given the dataset's mixed provenance.

We want to be explicit here: we did not apply any additional cleaning, deduplication, length filtering, or orthographic

normalization to this dataset beyond what the original release already includes. This was a deliberate choice rather than an oversight. Many published low-resource MT papers describe non-trivial preprocessing pipelines, including normalization of intra-word spacing artifacts, removal of duplicate or near-duplicate pairs, and filtering of misaligned sentences, and these steps have been shown elsewhere to produce meaningful chrF++ gains on their own, independent of any change to the model or training procedure. We did not perform any of this. As a direct consequence, the results in this paper reflect what a fully off-the-shelf, unmodified version of this dataset can produce when paired with a standard fine-tuning recipe, not the upper bound of what the dataset could support with proper curation, and not a result that should be benchmarked against systems that did include such preprocessing.

B. Base Model and Tokenization

We start from `facebook/nllb-200-distilled-600M`, the distilled 600M-parameter variant of NLLB-200. The tokenizer is configured with `spa_Latn` as the source language code and `quy_Latn` as the target, both of which are part of NLLB's native, pretrained language inventory, meaning the model already has some prior exposure to Quechua text before fine-tuning even begins. Each sentence pair is tokenized with a maximum sequence length of 128 tokens, with truncation and padding applied as needed. This length was chosen because the corpus consists predominantly of single, short-to-medium length sentences rather than long paragraphs, and 128 tokens comfortably covers the large majority of examples without excessive padding overhead.

C. Fine-Tuning Setup

Training was performed using the Hugging Face `Seq2SeqTrainer`. Table I summarizes the hyperparameters used. We used a learning rate of 3e-5 with the default AdamW optimizer settings, a per-device batch size of 4 combined with gradient accumulation over 4 steps, giving an effective batch size of 16, and trained under fp16 mixed precision for up to 10 epochs. Evaluation and checkpointing were performed at the end of every epoch, and early stopping was triggered if validation loss failed to improve for two consecutive epochs. We selected the best checkpoint by validation loss rather than by a generation-based metric such as chrF++, mainly for practical reasons, since computing generation metrics at every epoch would have added significant overhead to an already multi-hour training run on a single GPU. The best checkpoint, reached at step 57,798, had a validation loss of 0.2534.

All experiments were run on a single NVIDIA RTX PRO 6000 Blackwell Server Edition GPU with 102 GB of memory, accessed through a Google Colab instance. This particular GPU tier was chosen deliberately for its balance between cost and the compute the 600M model actually requires; a larger or more expensive instance was not necessary at this model scale, and we wanted the recipe to remain reproducible for researchers and small teams without access to dedicated compute clusters. The full fine-tuning run, including evaluation at each epoch, took between two and a half and three hours.

TABLE I
FINE-TUNING HYPERPARAMETERS

Hyperparameter	Value
Base model	NLLB-200-distilled-600M
Learning rate	3e-5
Per-device batch size	4
Gradient accumulation steps	4
Effective batch size	16
Max sequence length	128
Epochs (max / early stop)	10 / patience 2
Precision	fp16
Weight decay	0.01
Model selection metric	Validation loss
Best checkpoint (step)	57,798
Best validation loss	0.2534

D. Evaluation Protocol

We report two automatic metrics computed with `sacrebleu`: BLEU, the standard n-gram precision metric for machine translation, and chrF++ with word order 2, a character n-gram based metric that also incorporates word n-grams up to order 2. We rely more heavily on chrF++ throughout this paper because BLEU is known to penalize morphologically rich languages disproportionately: a translation that is semantically correct but differs from the reference in a single suffix, a common and often grammatically valid occurrence in Quechua, can fail to match any n-gram in the reference at all, even though the meaning conveyed is essentially identical. chrF++, by operating at the character level, is more forgiving of this kind of surface variation while still penalizing genuine translation errors, which makes it the more widely accepted metric in the AmericasNLP shared task series specifically because of this property.

Generation was performed with beam search using a beam size of 4 and a maximum output length of 128 tokens, matching the maximum input length used during tokenization. Two evaluation sets are used. The first is the held-out test split of the SomosNLP dataset itself, 12,843 examples drawn from the same distribution and collection process as the training data. The second is the official AmericasNLP 2024 development set for the Spanish-to-Quechua direction [3], downloaded directly from the shared task's public GitHub repository, containing 994 valid sentence pairs after filtering out empty lines. This second set was never seen during training and was constructed independently by a different research team for a different purpose, which makes it, at least in principle, a more meaningful test of generalization than the in-distribution test split, though as discussed in Section VI this independence turns out to be more limited than it might first appear.

V. Results

A. SomosNLP Test Set

On the 12,843-example test split of the SomosNLP dataset, the fine-tuned model achieves a BLEU score of 17.67 and a chrF++ score of 42.68. Given the morphological mismatch issues discussed above, the chrF++ score is the more informative figure here. A score in the low forties indicates the model is producing translations that are reasonably close to the references at the character level for a meaningful share of the test set, though clearly far from perfect, and not in a range that would be considered production quality for a high-resource language pair.

B. AmericasNLP 2024 Development Set

On the official AmericasNLP 2024 development set, the same model, with no further fine-tuning or adaptation, scores 51.88 chrF++ and 10.89 BLEU. Table II places this result alongside figures previously reported for the same benchmark by the BSC team [3], including the standard Helsinki and Sheffield baselines, a Google Translate reference point, and untuned and fine-tuned NLLB systems at 1.3B and 3.3B scale.

TABLE II
chrF++ and BLEU on the AmericasNLP 2024 Spanish→Quechua development set

System	chrF++	BLEU
Helsinki baseline [6]	28.78	–
Sheffield baseline [7]	30.22	–
Google Translate	34.86	3.23
NLLB-1.3B (no fine-tuning)	24.97	1.95
NLLB-1.3B + fine-tuning [3]	36.27	3.77
NLLB-3.3B + LoRA [3]	36.40	4.07
Ours: NLLB-600M + fine-tuning	**51.88**	**10.89**

Taken at face value, our 600M model outperforms every system in Table II, including the largest, most carefully tuned one. We want to be very direct about how this result should and should not be interpreted, since it is the single most striking number in this paper and also the easiest to misread. It does not mean our system is state of the art, and it should not be compared head-to-head with the BSC results, or with any industry-grade translation system, as if it were measuring the same thing under the same conditions. The remainder of this section, and Section VI in particular, explains why.

VI. Discussion

A. Why This Is Not a Fair Comparison to State-of-the-Art Systems

The gap between our score on the AmericasNLP dev set and every previously reported result is large enough that it would be irresponsible to present it without qualification. We believe the most likely explanation is not that our training recipe is fundamentally better, but that our evaluation set is, in effect, less independent from our training data than the comparison table suggests. Both the SomosNLP corpus and the data underlying the AmericasNLP shared task originate from overlapping corners of the Peruvian Quechua NLP community, share a similar Ayacucho-Chanka register, and consist predominantly of short, encyclopedic or conversational sentences rather than long-form, technical, or domain-specific text. Fine-tuning directly on SomosNLP data likely gives our model a stylistic and lexical advantage on the AmericasNLP dev set that would not be expected to transfer to genuinely out-of-domain text, legal documents, spontaneous speech, social media, or specialized technical content, none of which is well represented in either corpus.

On top of this, and just as important, the SomosNLP training data was used completely unmodified, with no cleaning, deduplication, or quality filtering, as described in Section IV. Systems like the BSC submission, by contrast, are typically built on top of data pipelines that include at least some normalization and filtering, and other recent work on this language pair has shown that orthographic normalization alone, addressing issues such as inconsistent intra-word spacing, can produce several points of chrF++ improvement independent of any change to the model. This means that even setting domain overlap aside entirely, our number and the BSC number are not measuring the output of comparable systems: one reflects a model trained on cleaned, curated data, the other reflects a model trained on data exactly as a community hackathon happened to produce it. For this reason, we do not present our result as evidence that a 600M model is competitive with, let alone superior to, a 3.3B model under matched conditions, and we caution strongly against citing the 51.88 chrF++ figure as a state-of-the-art result for Spanish-Quechua translation without these caveats attached.

B. Qualitative Error Analysis

To understand what the headline numbers actually look like in practice, we manually inspected a sample of model outputs on the AmericasNLP dev set. Table III shows representative cases.

TABLE III
Sample translations from the AmericasNLP 2024 dev set

ES: Solo dura una semana. **PRED:** Huk semanam. **REF:** Huk punchawkunallam kan
ES: Vamos a la cabaña en Minnesota. **PRED:** Minnesota sutiyuq wasiykuman riyku. **REF:** Minnesotapi chukllaman hakuchik
ES: Rudolph Anderson fue parte de tres U2. **PRED:** Rudolph Anderson sutiyoq runam kimsa U2 grupokunapi karqa. **REF:** Rudolph Andersonqa Kimsa U2 nisqapi karqa.
ES: Nuestras acciones les impiden mejorar. **PRED:** Imapas ruwasqanchikqa manam yanapanchu aswan allin kananku-paqqa. **REF:** Payqa papilta rachkaq hinaspa aquman hinaq, ninata rataykachiq hinaspa aqu sansapi kañaq, hinaspan kaynata sansata kuyuchiq.

Short, structurally simple sentences, the first two examples, are translated in a way that is semantically reasonable even when the surface realization differs noticeably from the reference, which chrF++ tolerates relatively well given its character-level matching. The third example illustrates a recurring pattern: the reference uses a more compressed

grammatical construction than our model's output, which is grammatically valid Quechua but more verbose than necessary, a stylistic mismatch rather than an outright error, though one that automatic metrics still penalize to some degree. The fourth example is the most concerning, and frankly the most useful one to include here: the Quechua reference bears no discernible relationship to the Spanish source sentence at all, describing what appears to be an entirely different scene. This is almost certainly a misalignment in the original AmericasNLP data rather than a failure on the model's part, and it is exactly the kind of issue that a data cleaning pass would catch and that we did not attempt to catch in this work. We include it deliberately, as a concrete illustration of why uncleaned, unverified data should not be trusted to produce a reliable automatic score, in either direction; this particular example would hurt our chrF++ number through no fault of the model, but other similarly noisy examples could just as easily inflate it.

C. Tokenization as a Remaining Bottleneck

Beyond data quality, the error patterns we observe are also consistent with the tokenization limitations discussed in Section III. Quechua's long suffix chains are split by NLLB's shared SentencePiece vocabulary in ways that frequently do not align with actual morpheme boundaries, since that vocabulary was optimized jointly across more than 200 typologically diverse languages rather than tailored to Quechua specifically. This likely makes it harder for the model to learn productive, generalizable affixation patterns from a training set of this size, and it offers one plausible explanation for why even well-translated outputs sometimes differ from references in their choice of suffix or in how compactly a given meaning is expressed. This is consistent with the broader argument made by Ortega et al. [8] for polysynthetic languages generally, and we suspect that a morphology-aware tokenizer, or at minimum a Quechua-specific subword vocabulary trained from scratch on a larger and more representative corpus, would meaningfully improve results independent of any change in model scale.

VII. Limitations

We want to state the limitations of this study plainly, in one place, rather than leaving them scattered across the discussion.

First, and most importantly, the training data was used without any cleaning, deduplication, filtering, or orthographic normalization. This is a deliberate methodological choice intended to measure performance under realistic, low-effort conditions, but it also means our results are not directly comparable to systems built on curated data pipelines, and they should not be cited as a benchmark for what is achievable with this dataset under better preprocessing.

Second, our strongest result, the AmericasNLP dev set score, is very likely inflated by domain and source overlap with our training data, as discussed at length in Section VI. We do not consider this number evidence of state-of-the-art performance, and we explicitly discourage readers from treating it as such or from using it to claim superiority over larger, more carefully evaluated systems like the BSC submission.

Third, we evaluate using automatic metrics only. BLEU and chrF++, even the more morphology-tolerant chrF++, are known to correlate imperfectly with human judgments of fluency and adequacy, particularly for agglutinative languages where multiple surface forms can express the same meaning equally correctly. No human evaluation was performed in this work.

Fourth, both of our evaluation sets are dominated by the Ayacucho-Chanka variety of Quechua. We make no claims about how this model performs on other Quechua varieties, such as Cusco-Collao or Bolivian Quechua, which differ enough linguistically that we would expect meaningfully worse performance without additional variety-specific data.

Finally, model selection during training was based on validation loss rather than a generation-based metric such as chrF++, for practical compute reasons. It is possible that a checkpoint other than the one with the lowest validation loss would have scored higher on either test set, and we did not explore this.

VIII. Conclusion

We fine-tuned the 600M distilled version of NLLB-200 on 102,000 Spanish-Quechua sentence pairs from the SomosNLP 2022 dataset, using the data exactly as released, and evaluated the resulting model on two test sets of different provenance. The model reached 42.68 chrF++ on its own in-domain test split and 51.88 chrF++ on the official AmericasNLP 2024 development set. The second number is, on the surface, higher than any previously published result for this benchmark, including a 3.3B parameter model fine-tuned with LoRA adapters, but we have argued at length that this should not be read as a state-of-the-art claim. The result is confounded by likely domain overlap between our uncleaned training corpus and the evaluation set, and it cannot be fairly compared to systems built on curated data or to commercial, industry-grade translation tools, which operate under very different data quality assumptions. Our qualitative analysis supports a more modest and, we think, more useful conclusion: that a fully fine-tuned, mid-sized 600M model performs respectably on short, structurally simple Quechua sentences, struggles more with longer or idiomatic constructions, and is held back as much by data quality and tokenization granularity as by raw model capacity.

For future work, we plan to build a proper, cleaned and deduplicated version of the SomosNLP corpus, with explicit filtering of misaligned pairs such as the one shown in Table III, and re-run this same fine-tuning recipe on top of it, to measure how much of the gap to genuinely state-of-the-art systems is attributable to data quality alone. We also intend to evaluate on a more clearly out-of-domain test set, ideally one drawn from a different register entirely, such as transcribed speech or government documents, to get a more honest measure of generalization than either of the two benchmarks used here can provide. Finally, we see real value in experimenting with

morphology-aware or Quechua-specific subword vocabularies, and in incorporating a small-scale human evaluation, both of which we believe are more promising directions for closing the remaining gap than simply scaling up the underlying model.

Code used for fine-tuning, evaluation, and the qualitative analysis reported here is available upon request.

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